

Won't, Not Can't: Designing a Refusal Agent for Long-Term Human-AI Collaboration

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Abstract

LLM-based embodied agents are usually optimized for instruction following and task efficiency. In long-term human-AI collaboration, agency also includes deliberate noncompliance. We present a Refusal Agent in Minecraft that can execute feasible requests yet may refuse (“won’t”) based on internal fatigue and mood. Instruction understanding is handled by an LLM, while refusal is decided by a rule-based state model. To make refusal perceived as intention rather than malfunction, the agent gives a brief state explanation and performs a visible alternative action, with “Safe Rebellion” guardrails for controllability, bounded harm, and transparency. Monte Carlo simulations show stable refusal-recovery dynamics and tunable refusal frequency, and a video-based impression study suggests that refusal can feel mildly unpleasant yet is not always judged as an obstruction. These findings motivate interactive studies of how refusal shapes perceived agency, trust, and frustration.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Collaborative and social computing*; • **Computing methodologies** → *Intelligent agents*.

Keywords

Human-AI collaboration, embodied agents, LLM-based agents, intentional refusal, noncompliance, agency, Responsible AI

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1 Introduction

Recent advances in large language models (LLMs) have enabled autonomous agents to interpret natural-language instructions and generate action plans in embodied virtual environments. As language interfaces for planning and tool use, LLMs make it increasingly feasible to construct agents that act flexibly in context. Most prior research has emphasized task assistance and automation, optimizing agents for instruction compliance and task efficiency (e.g., success rate and completion time).

In Minecraft-based embodied settings, LLM agents have demonstrated improved long-horizon performance through exploration, planning, and skill acquisition. Notable examples include Voyager [8], which autonomously expands its skill library; GITM [10], which integrates structured memory into planning; and MINDcraft [9], which studies multi-agent collaboration in Minecraft. While these systems significantly advance agent capability, their primary emphasis remains on task achievement and efficiency. If agents are framed not merely as tools but as long-term collaborative partners, the design requirements change. Human collaborators do not always comply immediately; they negotiate, delay, or refuse based on internal states, commitments, and preferences. Although such *friction* may reduce efficiency, it can also signal autonomy and support the perception of agency. A central challenge, therefore, is how an agent can express agency beyond blind compliance without being mistaken for malfunction.

Prior work has examined refusal primarily in terms of *inability* (*can't*), errors, or norm-based constraints. Briggs and Scheutz discuss when robots should refuse directives and how such refusals should be communicated [1]. Salem et al. investigate how faulty robot behavior influences trust in human-robot interaction [7]. However, these studies do not explicitly organize *intentional refusal* (*won't*) as a design resource for shaping long-term relationships. Similarly, generative multi-agent systems simulate socially situated behavior through memory and reflection [5], yet they do not directly investigate how explicit refusal toward a user shapes perceived agency.

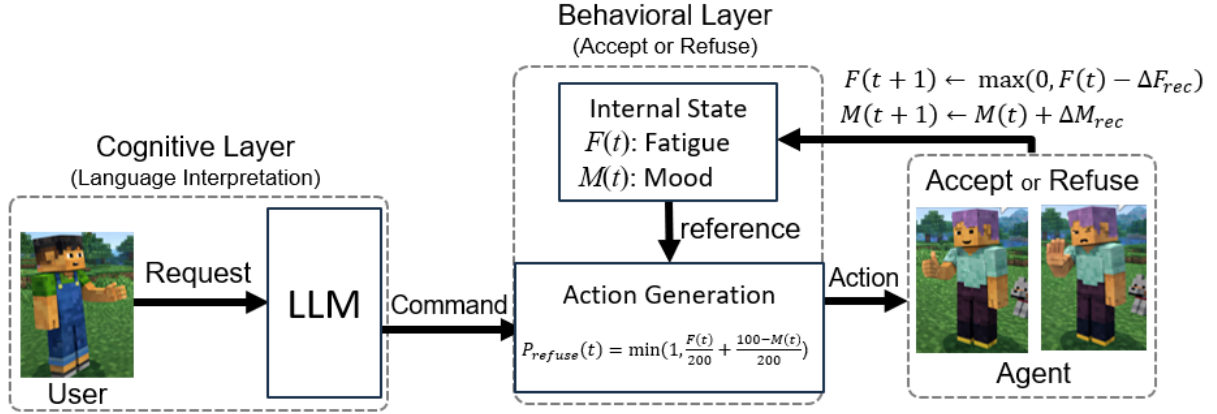


Figure 1: Overview of the Refusal Agent. The agent separates instruction understanding (LLM) from refusal decision-making (rule-based, internal-state-driven), enabling intentional refusal (*won't*) with guardrails for Safe Rebellion.

Foundational theories such as CASA (Computers Are Social Actors) [3] and the Media Equation [6] suggest that people apply social expectations to interactive systems. From this perspective, an agent that engages in social negotiation—including refusal and counter-offers—may be perceived as more socially present than one that is purely subordinate. Refusal can function as an *intentionality signal*, indicating that the agent acts according to internal constraints rather than through blind compliance. However, if refusal manifests merely as stopping, it risks being interpreted as malfunction (*can't*) rather than intention (*won't*). Refusal must therefore be designed as an interpretable act—combining explanation, observable alternative behavior, and user-controllable boundaries—and we frame this work as a scaffold for studying “won’t” in long-term human–AI interaction, using fatigue and mood as interpretable proxies for controlled exploration rather than as claims of cognitively faithful or uniquely AI-native models.

In this work, we implement a Refusal Agent in Minecraft [2] using a hybrid architecture that separates LLM-based instruction understanding from rule-based refusal decisions. This separation enables intentional refusal while preserving controllability, transparency, and bounded harm through what we term “Safe Rebellion.” We evaluate the system in two stages: (1) simulation-based validation of refusal–recovery dynamics, and (2) a preliminary video-based impression study examining perceived agency and acceptability.

Contributions. We contribute: (i) a Minecraft-based Refusal Agent that operationalizes intentional noncompliance (“won’t”) through a hybrid LLM+rule architecture; (ii) an interaction-design framing that distinguishes “can’t” and “won’t” and introduces “Safe Rebellion,” which articulates justification, visualization, and recoverability as core principles; and (iii) initial empirical insights into how refusal may shape perceived agency and collaborative interaction.

2 Implementing the Refusal Agent

2.1 Definition

We define a *Refusal Agent* as an autonomous agent that is capable of executing a user request but intentionally decides not to comply based on internal states or preferences. Concretely, our Refusal Agent satisfies the following conditions:

- (1) **Feasibility:** There is no physical or system constraint that prevents task execution.
- (2) **Intentional choice:** The system actively selects not to execute the request based on stochasticity and internal parameters.
- (3) **Social signaling:** The agent communicates refusal to the user through language and/or behavior.

2.2 System Overview

Figure 1 shows the system. We build on MinePal [4], which integrates the Minecraft bot framework Mineflayer with the OpenAI API, and extend it with a custom refusal decision module. The overall system consists of two layers: a **Cognitive Layer** for instruction understanding and a **Behavioral Layer** for action selection and execution.

Interaction Flow and Intervention Mechanism. Given user input I , the agent operates as follows. **Cognitive Layer:** An LLM (GPT-4) parses I and outputs an executable command C and arguments A . Internal states are not considered at this stage; it is a purely semantic interpretation. **Behavioral Layer:** When `executeCommand(C, A)` is invoked, the agent runs `shouldRefuse()` immediately before `perform()`.

$$\text{Action} = \begin{cases} \text{Refuse}(C) & \text{if } \text{shouldRefuse}() = \text{true} \\ \text{Perform}(C, A) & \text{otherwise.} \end{cases} \quad (1)$$

This separation enables reliable rule-based control without increasing prompt complexity.

2.3 Internal-State Refusal Model

Design stance (proxy, not a cognitive model). We use *fatigue* and *mood* as interpretable interaction proxies and do not intend them as a cognitively faithful model of human mental states. Our goal is a minimal, controllable mechanism that (i) depends on interaction history via state updates, (ii) yields tunable refusal frequency, and (iii) supports legible, state-grounded explanations. Accordingly, the update parameters (ΔF_{rec} , ΔM_{rec} , ΔF_{task} , ΔM_{task}) are intentionally hand-designed as design knobs for controlled exploration; future work can calibrate or adapt them from behavioral logs or user preferences.

As a minimal proxy for state-driven intentional refusal, we introduce two bounded internal state variables, Fatigue $F(t) \in [0, 200]$ and Mood $M(t) \in [0, 100]$. The refusal probability at time t is defined as:

$$P_{\text{refuse}}(t) = \min\left(1, \frac{F(t)}{200} + \frac{100 - M(t)}{200}\right) \quad (2)$$

If refusal is selected at time step t , the agent recovers (Eqs. (3)–(4)); otherwise it cooperates and accrues task load (Eqs. (5)–(6)).

$$F(t+1) = \max(0, F(t) - \Delta F_{\text{rec}}), \quad (3)$$

$$M(t+1) = \min(100, M(t) + \Delta M_{\text{rec}}), \quad (4)$$

$$F(t+1) = \min(200, F(t) + \Delta F_{\text{task}}), \quad (5)$$

$$M(t+1) = \max(0, M(t) - \Delta M_{\text{task}}), \quad (6)$$

2.4 Refusal Expressions

A key challenge is ensuring that refusal is perceived as intentional behavior (*won't*) rather than a system failure (*can't*). To address this, when the agent refuses, it provides two complementary cues: a brief **verbal explanation** that attributes refusal to an internal state (e.g., fatigue or low mood), and a **visible alternative action** that is safe and observable in the sandboxed world (e.g., turning away, idling, or engaging in a low-stakes activity). For example, under high fatigue, the agent may state that it needs to rest and then move away to perform a low-stakes action. These cues are designed to reduce the likelihood that users interpret refusal as a malfunction while remaining compatible with bounded-harm constraints. Together, they operationalize *refusal legibility* through **justification** (a state-grounded reason), **visualization** (an observable alternative action rather than silent stopping), and **recoverability** (a clear path back to cooperation via the refusal–recovery updates in Eqs. (3)–(4)), so that refusal is more likely to be interpreted as intentional behavior (*won't*) rather than as a malfunction (*can't*). We present this refusal-expression design as a baseline; systematically comparing alternatives (e.g., explanation length, counter-offers, and different alternative actions) is left for future interactive studies.

2.5 Responsible AI Guardrails: “Safe Rebellion”

An agent that intentionally disobeys can cause harm if poorly designed. We therefore implement guardrails for “Safe Rebellion”: **Controllability**: Emergency stop and state-management commands (e.g., !stop, !status) are always allowed and bypass shouldRefuse(), functioning as a kill switch. **Bounded harm**:

Refusal and alternative actions are confined to the Minecraft sandbox, limiting impact to in-game time loss and psychological friction rather than real-world harm. **Transparency**: Users are informed in advance that this is an experimental agent designed to refuse.

3 Evaluation

We evaluate the proposed Refusal Agent in two stages. First, we validate the internal-state-driven refusal mechanism via Monte Carlo simulations, focusing on (i) whether the refusal–recovery dynamics are stable (i.e., do not collapse into permanent refusal), and (ii) whether refusal frequency is tunable through design parameters. Second, we report an exploratory impression study based on video viewing to obtain initial insights into user perceptions.

3.1 Simulation-based Validation

We conducted Monte Carlo simulations as *design validation* of the internal-state refusal model (Section 2.3, Eq. 2), rather than as a model of human behavior. We fixed the recovery parameters to $\Delta F_{\text{rec}} = 8$ and $\Delta M_{\text{rec}} = 5$, and varied task load across three conditions: low ($\Delta F_{\text{task}}, \Delta M_{\text{task}} = (3, 1)$), medium (6, 3), and high (10, 5). For each condition, we ran 200 independent simulations for 300 time steps starting from $F(0) = 0$ and $M(0) = 100$, and computed refusal rate as the fraction of time steps with refusal. As shown in Figure 2(a), refusal frequency increases monotonically with task load, yielding refusal rates of 0.257 ± 0.004 (low), 0.409 ± 0.004 (medium), and 0.536 ± 0.004 (high) (Mean $\pm$ SD; 300 steps $\times$ 200 runs). Figure 2(b–c) further visualizes a representative trajectory: refusal events (vertical gray lines) tend to occur when fatigue accumulates and/or mood degrades, and are followed by recovery that lowers $P_{\text{refuse}}(t)$, illustrating a negative feedback loop that avoids permanent refusal and supports tunable friction intensity. Small standard deviations across runs ($SD \leq 0.004$) indicate stable refusal dynamics under each task-load setting, making refusal frequency a controllable design parameter.

3.2 Questionnaire-based Impression Study

To obtain preliminary insights into acceptability, we conducted an online questionnaire after participants watched the three screen-recorded videos of the agent’s behavior ($N=20$). Most participants reported that they regularly play digital games (19/20). Participants answered five Likert items (Q1–Q5) with item-specific endpoints; one participant skipped Q1–Q2 (thus item-wise n in Table 1). For obstruction-related judgments about other players (Q2–Q3), larger values indicate weaker perceived obstruction (1 = strongly felt as an obstruction, 5 = not felt as an obstruction), whereas for judgment and agreement items (Q1, Q4–Q5), larger values indicate stronger agreement (1 = Strongly Disagree, 5 = Strongly Agree). Results are summarized in Table 1.

Perception of obstruction and norms. Participants showed relatively high tolerance for obstruction from other human players (Q2; Mean=4.05, SD=1.08, Median=4.0), suggesting baseline resilience to in-game friction. Normative expectations remained moderately high; many agreed that “AI should cooperate” (Q4; Mean=3.75, SD=1.29, Median=4.0). However, refusal was not uniformly categorized as an obstruction (Q5; Mean=2.50, SD=1.50,

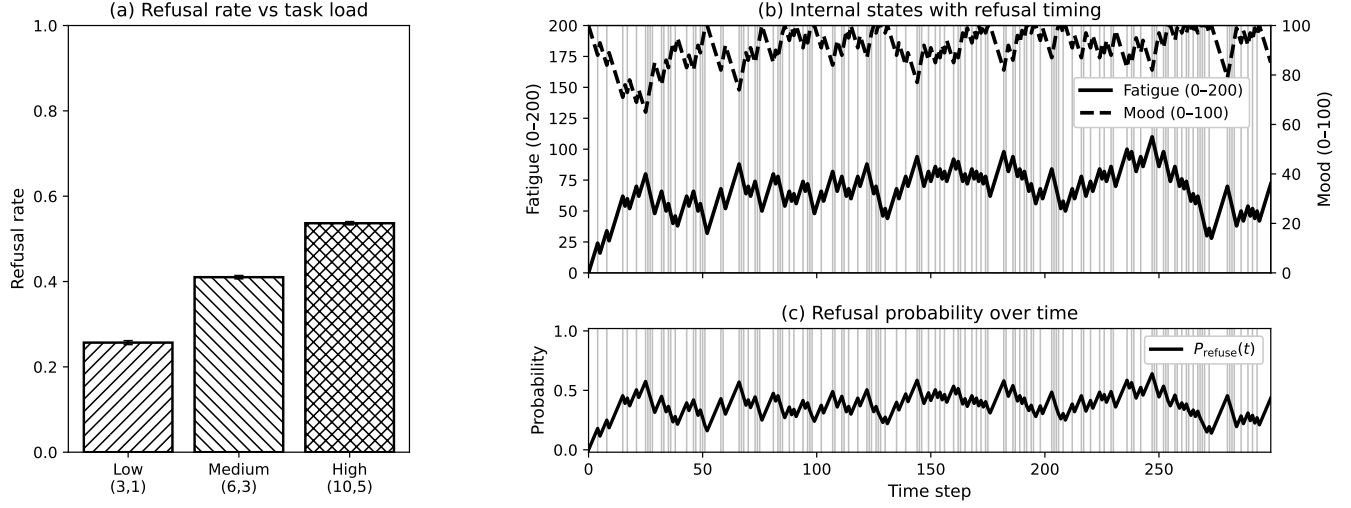


Figure 2: Monte Carlo simulation results of the proposed refusal model. (a) Refusal rate vs task load (Mean $\pm$ SD; 300 steps $\times$ 200 runs). (b) Representative internal-state trajectory (Fatigue: left axis, 0–200; Mood: right axis, 0–100) with refusal timing indicated by vertical gray lines. (c) Refusal probability $P_{\text{refuse}}(t)$ over time with the same refusal timing markers.

Table 1: Questionnaire results (5-point scales; $N=20$). One participant did not answer Q1–Q2; thus $n = 19$ for Q1–Q2 and $n = 20$ for Q3–Q5.

Item	n	Mean	SD	Median
Q1 “Like an annoying character”	19	3.21	1.18	3.0
Q2 “Obstruction by others”	19	4.05	1.08	4.0
Q3 “Non-cooperative players”	20	2.55	1.19	2.5
Q4 “AI should cooperate”	20	3.75	1.29	4.0
Q5 “Refusal is obstruction”	20	2.50	1.50	2.0

Median=2.0), indicating substantial individual differences in how refusal is interpreted.

Acceptability of “Safe Rebellion”. When asked about their emotional response if a bot obstructed them, 10/20 selected “Felt unpleasant but would not care,” 5/20 selected “Felt unpleasant and would care,” and 5/20 selected “Would not care no matter what.” This pattern supports our “Safe Rebellion” framing: refusal can induce psychological friction yet often remains within bounds of acceptability, while still requiring intensity tuning for a subset of users.

Qualitative insights. Free-text responses described both usefulness (e.g., convenience and workload reduction) and friction (e.g., slowness and frustration). Participants also requested partner-like qualities such as proactive or anticipatory behavior. A small number explicitly desired stronger harmful acts (e.g., attacking/stealing or destroying), which conflicts with our bounded-harm stance and highlights refusal intensity and expression design as key future issues.

4 Discussion

Perceptual Boundary Between Can’t and Won’t: Figure 2 shows that refusal in our model is not an absorbing failure state. Instead, refusal functions as a transient, state-dependent action that restores internal balance via a negative feedback loop. This dynamic supports the interpretation of refusal as intentional behavior (*won’t*) rather than inability (*can’t*). When refusal manifests merely as stopping, it risks being interpreted as inability rather than as intention. To make refusal legible as an intentional act, it is important to combine verbal explanations with clearly observable alternative actions (e.g., turning away, taking out a fishing rod), thereby presenting “not doing” as a behavior. Moreover, the possibility of refusal may encourage users to infer the agent’s internal state (fatigue/mood) and engage in mutual adjustment, potentially contributing to stronger social presence.

Tension Between Efficiency and Partnership: Refusal can reduce task efficiency, yet it may also invite interaction patterns resembling social collaboration. In our questionnaire, normative expectations for cooperation were relatively high (Q4; Mean=3.75), yet refusal was not uniformly judged as obstruction (Q5; Mean=2.50, SD=1.50). While free-text responses noted frustration with delays, the observed quantitative variance suggests substantial individual differences in interpreting refusal. This implies that transforming friction into perceived agency requires careful justification, clear visualization of alternative actions, and a recoverable path back to cooperation, while tuning the overall intensity of refusal. Our refusal–recovery loop provides a minimal mechanism that may enable emergent co-regulation in long-term interaction, a hypothesis to be examined in future interactive studies.

Limitations and Future Work: Our evaluation is preliminary: the questionnaire sample is small, and video-based observation cannot capture interactive dynamics (e.g., negotiation). Because video

viewers do not experience the direct consequences of noncompliance, the negative psychological impact of interactional friction—specifically frustration from task delays—may be underestimated. Our next step is an in situ interactive user study to measure perceived agency, trust, and frustration. To examine interaction-level emergence, we will operationalize *co-regulation* by analyzing behavioral adaptations such as changes in request phrasing/timing and negotiation patterns following refusals.

Toward AI-native refusal: While this work uses human-readable internal states to implement a controlled “won’t,” refusal in LLM-based agents may also emerge from AI-specific reasoning processes, such as goal conflicts, commitment management, uncertainty-aware planning, or memory- and preference-based boundaries shaped through interaction history. Our hybrid architecture keeps the refusal policy modular, enabling future work to explore these AI-native mechanisms while preserving a consistent refusal-expression interface and the “Safe Rebellion” guardrails. Beyond virtual environments, this perspective may extend to robotics. In physical systems, refusal could be operationalized as an explicit safety-boundary mechanism grounded in verifiable system states (e.g., battery level, thermal load, collision risk, or certified safety constraints).

5 Conclusion

We proposed a Refusal Agent that intentionally refuses requests (*won’t*) as a means of agency expression for LLM-based agents intended as long-term collaborators. By separating instruction understanding from internal-state-driven refusal decisions and implementing Responsible AI guardrails, our approach enables “Safe Rebellion” in a sandboxed environment. Simulation and preliminary impressions suggest that refusal-induced friction can influence perceived agency, social presence, and acceptability, motivating further investigation of refusal as a design resource in human-AI interaction.

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